

Linux - Linux Cheat Sheet Atlassian

Linux cheat sheet to support Atlassian server/DC products

Linux Cheat Sheet for Supporting Atlassian Server/Data Center Products

Applies to:

- Atlassian
- Jira
- Confluence
- Bitbucket
- Bamboo

Useful for:

- JVM troubleshooting
- Cluster diagnostics
- Database/network analysis
- Disk/memory pressure
- Reverse proxy debugging
- GC and thread investigations
- Slow node / split-brain / LB failures

1. System Health Quick Checks

CPU / Load

```
Bash
uptime
top
htop
mpstat -P ALL 1
vmstat 1
sar -t 1 10
```

Find top CPU consumers

```
Bash
ps -eo pid,ppid,cmd,%mem,%cpu --sort=-%cpu | head -20
```

Per-thread CPU (very important for Jira/Confluence)

```
Bash
top -H -p <PID>
```

Convert thread ID to hex for thread dump matching:

```
Bash
printf '%x\n' 12345
```

2. Memory Diagnostics

Overall memory

```
Bash
free -h
cat /proc/meminfo
vmstat -s
```

Detect swapping

```
Bash
vmstat 1
sar -H 1 10
```

Process memory

```
Bash
ps -eo pid,cmd,%mem,rss,vsz --sort=-rss | head
```

OOM killer events

```
Bash
dmesg -f | grep -i oom
journalctl -k | grep -i oom
```

3. Java/JVM Troubleshooting

Find Java processes

```
Bash
jps -tv
ps -ef | grep java
```

Thread dumps

Safe method

```
Bash
jstack -l <PID> > threaddump.txt
```

Fast emergency method

```
Bash
kill -3 <PID>
```

Thread dump appears in:

- catalina.out
- Jira logs
- Confluence logs

Heap info

```
Bash
jcmd <PID> GC.heap_info
jmap -heap <PID>
```

Heap dump

```
Bash
jmap -dump:live,format=b,file=heap.hprof <PID>
```

Or:

```
Bash
jcmd <PID> GC.heap_dump heap.hprof
```

GC diagnostics

Enable GC logs

JDK11+:

```
Bash
-Xlog:gc*:file=/var/log/jira/gc.log:time,uptime,level,tags
```

Monitor live GC

```
Bash
jstat -gcutil <PID> 1s
```

4. Atlassian Log Analysis

Tail logs live

```
Bash
tail -f atlassian-jira.log
tail -f catalina.out
tail -f atlassian-confluence.log
```

Search errors

```
Bash
grep -Ri "ERROR" .
grep -Ri "Exception" .
grep -Ri "OutOfMemory" .
```

Multi-pattern search

```
Bash
grep -E "ERROR|FATAL|OOM|deadlock|timeout" atlassian-jira.log
```

Context around errors

```
Bash
grep -n -A 20 -B 20 "ERROR" atlassian-jira.log
```

Paginated viewing without line splitting

```
Bash
grep -E "ERROR|WARN" atlassian-jira.log | less -S
```

Options:

- -s = do not wrap long lines
- Space = next page
- /text = search
- G = end
- g = beginning

5. Large Log File Operations

Biggest files

```
Bash
du -sh *
du -sh | sort -rh | head -30
```

Live growing files

```
Bash
watch -n 2 'ls -lh *.log'
```

Find logs modified recently

```
Bash
find . -name "*.log" -mtime -1
```

Compress old logs

```
Bash
gzip *.log
```

6. Network Diagnostics

Listening ports

```
Bash
ss -ltnp
netstat -ltnp
```

Established connections

```
Bash
ss -ant
```

Count connections per IP

```
Bash
ss -ant | awk '{print $5}' | cut -d: -f1 | sort | uniq -c | sort -nr
```

Test HTTP response

```
Bash
curl -I http://localhost:8080
curl -vk https://jira.company.com
```

DNS checks

```
Bash
dig jira.company.com
nslookup jira.company.com
```

Packet capture

```
Bash
tcpdump -i any host <IP>
```

7. Disk / IO Troubleshooting

Disk usage

```
Bash
df -h
du -sh *
```

IO bottlenecks

```
Bash
iostat -xz 1
iotop
```

Open deleted files (common hidden disk leak)

```
Bash
lsof | grep deleted
```

8. Database Connectivity

Test PostgreSQL

```
Bash
psql -h dbhost -U jirauser jira
```

Active DB sessions

PostgreSQL:

```
SQL
SELECT pid, username, state, query
FROM pg_stat_activity;
```

DB connection saturation

```
Bash
netstat -an | grep 5432 | wc -l
```

9. Cluster / Data Center Diagnostics

Verify cluster traffic

```
Bash
ss -ant | grep 40001
```

Typical ports:

- Jira Hazelcast
- Confluence clustering
- Synchrony

Detect split brain indicators

```
Bash
grep -Ri "split" atlassian-jira.log
grep -Ri "hazelcast" atlassian-jira.log
grep -Ri "ClusterJoinManager" .
```

Verify shared home mount

```
Bash
mount
df -h
touch /shared/testfile
```

10. Reverse Proxy / Load Balancer

Nginx validation

```
Bash
nginx -t
```

Apache validation

```
Bash
apachectl configtest
```

Verify headers

```
Bash
curl -i -H "X-Forwarded-Proto: https" http://localhost:8080
```

Sticky session checks

```
Bash
grep JSESSIONID access.log
```

11. Service Management

systemd

```
Bash
systemctl status jira
systemctl restart jira
systemctl stop jira
journalctl -u jira -f
```

Process tree

```
Bash
pstree -p
```

12. Useful One-Liners

Top 20 slowest Jira requests

```
Bash
grep "REQUEST" atlassian-jira.log | sort -kNF
```

Find deadlocks

```
Bash
grep -Ri deadlock .
```

Detect stuck threads

```
Bash
grep -Ri "WAITING" threaddump.txt
grep -Ri "BLOCKED" threaddump.txt
```

Find repeated errors

```
Bash
grep ERROR atlassian-jira.log | sort | uniq -c | sort -nr | head
```

Find plugin-related failures

```
Bash
grep -Ri "plugin" atlassian-jira.log
```

13. Critical Jira/Confluence Directories

Jira

Typical:
`/opt/atlassian/jira/`
`/var/atlassian/application-data/jira/`

- Important:
- logs/
 - plugins/
 - caches/
 - tmp/
 - export/
 - import/

Confluence

Typical:
`/opt/atlassian/confluence/`
`/var/atlassian/application-data/confluence/`

- Important:
- logs/
 - temp/
 - plugins-cache/
 - bundled-plugins/
 - shared-home/

14. Emergency Triage Flow

Jira Node Suddenly Slow

Step 1

Check:

```
Bash
top
vmstat 1
iostat -xz 1
```

Step 2

Check JVM:

```
Bash
jstat -gcutil <PID> 1s
top -H -p <PID>
```

Step 3

Capture evidence:

```
Bash
jstack -t <PID> > tdump.txt
```

Step 4

Check logs:

```
Bash
grep -E "ERROR|WARN|deadLock|timeout|OOM" atlassian-jira.log
```

Step 5

Check DB:

```
Bash
ss -ant | grep 5432
```

15. Essential Packages to Install

Debian/Ubuntu

```
Bash
apt install:
htop
iotop
sysstat
net-tools
tcpdump
lsof
jq
curl
wget
zip
unzip
dnsutils
openjdk-17-jdk
```

16. Extremely Useful Commands for Enterprise Jira DC

Correlate CPU thread to Java stack

```
Bash
top -H -p <PID>
printf '%x\n' <TID>
grep -i nid=0xHEX threaddump.txt
```

Monitor GC pauses in real time

```
Bash
tail -f gc.log | grep Pause
```

Detect excessive open files

```
Bash
lsof -p <PID> | wc -l
```

Limits:

```
Bash
ulimit -n
cat /proc/<PID>/limits
```

Identify busiest endpoints

Nginx:

```
Bash
awk '{print $7}' access.log | sort | uniq -c | sort -nr | head
```

17. Recommended Enterprise Tooling

System

- htop
- sysstat
- atop
- glances

JVM

- Eclipse MAT
- VisualVM
- JMC (Java Mission Control)

Logs

- ELK stack
- Splunk
- Grafana Loki

Monitoring

- Prometheus
- Grafana
- Zabbix

18. Fast Incident Commands Bundle

```
Bash
date
hostname
uptime
free -h
df -h
vmstat 1 5
iostat -xz 1 5
top -b -n1 | head -40
jps -lv
ss -s
```

19. Safe Production Practices

Avoid in peak hours unless necessary:

- heap dumps
- full GC forcing
- restarting cluster nodes
- aggressive grep on multi-GB logs
- recursive chmod/chown

Prefer:

```
Bash
nice
ionice
less
tail
```

20. High-Value Log Locations

Jira

```
logs/atlassian-jira.log
logs/catalina.out
logs/access_log.*
logs/gc.log
```

Confluence

```
logs/atlassian-confluence.log
logs/catalina.out
logs/synchrony-proxy.log
logs/gc.log
```

Linux

```
/var/log/messages
/var/log/syslog
/var/log/kern.log
journalctl
```

21. Recommended Atlassian References

- [Atlassian Support](#)
- [Atlassian Data Center Documentation](#)
- [Jira Data Center Docs](#)
- [Confluence Data Center Docs](#)